Active Observability of Chronological Carry Recurrences: Masking, Probing, and Sharp Intervention Costs

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Abstract

We investigate the recovery of hidden chronological sources from incomplete coefficient histories of the typed terminal-block CCR–k recurrence. A three-step masking construction admits distinct old histories with identical future outputs in the absence of further excitation, whereas rank–time controlled pulses can restore identifiability under explicit restrictions on fresh high-degree nuisance inputs. We reduce the full-row interrogation problem to additive finite-field experiments while retaining the nonresetting CCR recurrence. For a CCR–10 source family over $\mathbb{F}_{17}$, the minimum worst-case intervention count is two for adaptive pulse selection but three for predetermined schedules. Under a uniform prior and independent uniform bounded diagnostic nuisance, the minimum expected pulse count subject to a worst-case two-pulse limit is $44/25$, and the unrestricted three-slot expected optimum is $199/125$. Beyond four individually certified primes, an explicit family exhibits a strict worst-case gap for every prime $q \geq 41$: fixed intervention schedules require three pulses, whereas adaptive schedules require only two under the stated CCR–10 restricted-input contract, even when the prepared and later complete coefficient rows are revealed. All sharp constants refer to the specified source contracts; finite-state optimization methods are classical.

Keywords: chronological carry recurrence; active observability; partial observation; adaptive identification; finite fields; sharp intervention bounds.

1 Introduction

Chronological carry recurrences separate elementary source inputs from transported coefficient histories. Earlier CCR–3, CCR–4, and CCR–k studies developed the source reconstruction and connected-block formulation [1, 2, 3]; the separate arithmetic-refinement paper concerns p-adic resonance, ramification and density laws [4]. Here we examine a different question: whether a hidden old source can be identified with a limited number of active interventions despite unknown contemporaneous higher-order sources.

The distinction between adaptive and preset experiments is familiar in finite-state checking sequences [5], and observability with unmeasured inputs is a developed topic for nonlinear discrete-time systems [8]. Hence neither adaptive decision trees nor affine-output collision tests are claimed as new. Our focus is the specific source–time intervention normal form and the sharp pulse gap in an exact CCR recurrence.

Two features distinguish this inverse problem from reconstructing sources when complete chronological input data are given. Distinct old histories may lead to the same recent coefficient rows, so passive observation alone need not recover the discrepancy. Moreover, unknown higher-rank inputs can conceal the response to a controlled pulse. It is therefore essential to specify which source terms are allowed to vary, what the observer can see, and whether a diagnostic reading captures the information in a complete coefficient row.

2 Contributions

We connect (i) the CCR–k chronology and its finite masking horizon, (ii) rank–time-normalized controlled interrogation with bounded unknown high-degree sources, (iii) a sharp adaptive-versus-predetermined pulse count separation, and (iv) an exact worst-case versus expected pulse-cost comparison. Classical tools (triangular elimination, finite-field difference sets and Bellman recursion) are used explicitly and are not claimed as new in isolation.

3 Chronological recurrence and normalized readouts

Fix an integer p, a cutoff k, and write $C_{n,j} = [u^j]P_n(u)$. The typed source system is

$$P_n(u) = (1 + \ell_n u)P_{n-1}(u) + p(n-1)\ell_{n-1}u^2 P_{n-2}(u) + \sum_{r=3}^{k}(n-1)_{r-1}a_{r,n}u^r P_{n-r}(u) \pmod{u^{k+1}}, \tag{1}$$

where $\ell_n = (p+1)n - p$, $a_{3,n} = -p$, $P_0 = 1$ and $P_n = 0$ for $n < 0$. The falling factorial $(m)_r$ denotes $m(m-1)\cdots(m-r+1)$. The source inputs $a_{r,n}$ for $r \geq 4$ are treated as part of the explicit model, not automatically identified with arbitrary external events.

For times s and degrees j, let $R_{s,j}$ be the coefficient obtained from the three preceding complete observed rows by deleting the fresh higher-source summands in (1). The key observation is that current source rank r has a diagonal coefficient $(s-1)_{r-1}$, and the higher current sources can be solved recursively whenever these factors are units in the field. This lets us compare full-row collisions rather than appealing only to a heuristic scalar readout.

4 Passive masking and the need for active interrogation

The source separation theorem of CCR–k makes new rank-r sources triangular in the observed coefficient of degree r. Earlier source variables interact once their chronological blocks can occupy disjoint intervals. Starting at rank eight, two independent rank-four source blocks can contribute multiplicatively. In particular, the effect of an old rank-four source on the current rank-eight diagnostic may become visible only after a controlled rank-four pulse.

The three-lag baseline in (1) also gives a sharp form of persistent passive masking. Take two old histories that differ only at rank four, a prime field $\mathbb{F}_q$ with $q > n_0 + 3$, and let the fresh source of rank eight be fixed to zero during times n_0, n_0+1, n_0+2, while other high-rank sources can vary with the old history. The fresh-source map to a full output row is triangular, with nonzero diagonal. Fixing rank eight restricts the row image to an affine hyperplane. The two histories change the corresponding normal in the rank-four direction, and the two hyperplanes meet in q^{k-5} output rows. Three consecutive common rows can therefore be achieved. If all later high sources vanish, the common output rows perpetuate indefinitely because the remaining recurrence has three delays. This fact is not a claim that the system is unobservable under all future controlled inputs.

Under a known rank-four pulse at time $n_0 + 3$, with all other fresh high sources zero, the old source difference reappears at degree eight:

$$\Delta C_{n_0+3,8} = \alpha(n_0+2)_3(t-1)_3\Delta h \pmod{q}.$$

This calculation is a special case of the rank–time interrogation normal form below and motivates the bounded-nuisance experiment.

4.1 A sharp adaptive–predetermined separation in CCR–10

Let $\mathbb{F} = \mathbb{F}_{17}$ and use the typed terminal-block CCR–k recurrence with $(p, k, n_0, t) = (5, 10, 11, 4)$, where all identities are in $\mathbb{F}[u]/(u^{11})$. The only varying old source before time 11 is $h = a_{4,4}$ from $H = \{3, 5, 6, 11, 16\}$. At times $11, 12, 13$ all higher sources $r = 4, \dots, 10$ may depend on h and are chosen to yield the same three full coefficient rows for every h. The designated nonresetting pulse opportunities are $(14, 4), (15, 5), (16, 6)$, where a rank-$r$ pulse $\alpha_r \neq 0$ and independent diagnostic nuisance $z_r = a_{r+4,10+r} \in E = \{0, 1, 2, 3, 4\}$ are allowed. All fresh sources above rank $r + 4$ are unrestricted; all other fresh high sources are zero. At unused opportunities all fresh high sources are zero. The observer sees complete coefficient rows. Cost is the number of nonzero chosen pulses.

Write $B = C_{10,4}(h = 0)$, and let $R_{s,j}$ be the baseline obtained from the three previous observed full rows with fresh high sources zero. For $s_r = 10 + r$ define

$$w_r = (C_{s_r,r+4} - R_{s_r,r+4} - (s_r - 1)_{r-1}\alpha_r B)\,[(s_r - 1)_{r+3}]^{-1}.$$

The rank–time reduction from the original CCR recurrence gives

$$w_r = c\alpha_r h + z_r, \qquad c = (3)_3[(10)_4]^{-1} = 5 \pmod{17}. \tag{2}$$

The source–to–row triangular pivots $(s - 1)_{j-1}$ are units in $\mathbb{F}_{17}$ through time 16. Therefore any two histories with identical previous three full rows and the same w_r can match the *entire* next full row by choosing unrestricted sources of degrees $r + 5, \dots, 10$ successively.

Theorem 4.1 (Strict adaptive advantage with no reset)**.** *In this source and interrogation contract, the minimum worst-case number of active pulses is* 2 *for an adaptive controller but* 3 *for a controller whose pulse times and nonzero amplitudes must be predetermined without seeing intermediate coefficient observations. Both bounds are attained.*

Proof. At time 14 use $\alpha_4 = 14$, corresponding to normalized slope $a_4 = c\alpha_4 = 2$. The possible first normalized outputs $y = 2h + z_1$ leave candidate sets

y	$0, 1, 2$	5	$6, 7, 8, 9$	10	11	$12, 13, 14$	$15, 16$
H_y	$\{16\}$	$\{11\}$	$\{3, 11\}$	$\{3, 5\}$	$\{5\}$	$\{5, 6\}$	$\{6, 16\}$

with other residues impossible. If H_y is a singleton, stop. Otherwise pulse at time 15 with a normalized slope a_5 equal to 1 for $H_y = \{3, 11\}, \{6, 16\}$, equal to 3 for $H_y = \{3, 5\}$, and equal to 5 for $H_y = \{5, 6\}$. The actual rank-five amplitude is $\alpha_5 = 7a_5$ in $\mathbb{F}$. Since the nonzero pair differences, multiplied by the chosen slopes, are respectively $\pm 8, \pm 10, \pm 6, \pm 5$, none lies in $E - E = \{0, \pm 1, \pm 2, \pm 3, \pm 4\}$, hence the two normalized readouts uniquely determine h for every allowed nuisance history. The triangular completion argument makes the conclusion valid for full coefficient trajectories.

One pulse cannot identify all five h robustly: each generates five diagnostic residues, so $25 > 17$ prohibits pairwise disjoint one-pulse attainable-output sets. Thus the adaptive optimum is 2.

For the predetermined lower bound, note that $H - H = \mathbb{F}$. Set $D = (E - E) \setminus \{0\} = \{\pm 1, \pm 2, \pm 3, \pm 4\}$. At a nonzero slope a a difference d is unseparated precisely when $ad \in D$. The element $g = 3$ generates $\mathbb{F}^\times$; modulo the sign subgroup $\{\pm 1\}$, D has exponent classes $S = \{0, 1, 4, 6\}$ in $\mathbb{Z}/8\mathbb{Z}$. A direct calculation gives $S - S = \mathbb{Z}/8\mathbb{Z}$, so for any two nonzero slopes a, b, the sets $a^{-1}D$ and $b^{-1}D$ intersect. Every such nonzero difference occurs between two old values in H. Hence no predetermined two-pulse schedule separates all old histories; unrestricted higher sources can complete a common full trajectory.

Finally, the three predetermined slopes $(2, 1, 5)$ have $2^{-1}D \cap 1^{-1}D \cap 5^{-1}D = \varnothing$; at the three opportunities they correspond to amplitudes $(14, 7, 1)$. Thus three predetermined pulses suffice and are necessary. □

Remark 4.2 (Finite adaptive decision rule)**.** *For any finite old-source set S, nuisance set E, and normalized slope a, the posterior candidate set after observed y is $S_{a,y} = \{h \in S : y - ah \in E\}$. A finite-horizon minimax recursion over these candidate sets gives the exact adaptive worst-case pulse cost. The decision-tree principle itself is classical; the point here is its exact realization by the nonresetting typed CCR–k recurrence under the specified higher-source contract.*

5 Sharp expected intervention cost

The preceding sharp separation compares *worst-case* counts. We now consider the same three nonresetting opportunities, but assume that the old source is uniform on $H = \{3, 5, 6, 11, 16\}$ and each diagnostic nuisance z_i is independently uniform on $E = \{0, 1, 2, 3, 4\}$, conditioned only on an intervention being used. All other unknown high sources remain arbitrary and have no effect on the normalized diagnostics. The cost of a policy is its number of activated slots. Skipping an opportunity does not add an observation; the history still advances under the specified zero-source contract.

For a current candidate set $S \subseteq H$ define

$$S_{a,y} = \{h \in S : y - ah \in E\}, \qquad a \in \mathbb{F}_{17}^{\times}.$$

Writing $V_i(S)$ for minimum expected *remaining* pulse cost at opportunity $i \in \{14, 15, 16\}$ yields the exact finite recursion

$$V_i(S) = \min\left\{V_{i+1}(S), \quad 1 + \min_{a \in \mathbb{F}_{17}^{\times}} \sum_{y : S_{a,y} \neq \varnothing} \frac{|S_{a,y}|}{5|S|} V_{i+1}(S_{a,y})\right\}, \tag{3}$$

with $V_i(S) = 0$ for $|S| \leq 1$ and $V_{17}(S) = +\infty$ for $|S| > 1$. The same recursion with a restriction on the number of remaining pulses characterizes expected cost under a worst-case pulse budget. These are classical finite-state decision-tree recursions; here they are applied to the exact CCR–10 additive probe normal form.

Theorem 5.1 (Sharp worst-case/mean-cost tradeoff for CCR–10)**.** *Under the CCR–10 source contract of the preceding section and the specified independent uniform prior and diagnostic nuisances:*

1. *The minimum worst-case pulse count is 2, while the minimum expected count among strategies that* never *use more than two pulses is exactly $44/25 = 1.76$.*
2. *If all three opportunities may be used on exceptional trajectories, the minimum expected pulse count is exactly*

$$\boxed{V_{14}(H) = \frac{199}{125} = 1.592.}$$

This optimum is attained by a policy whose worst-case count is 3.

Proof. The robust worst-case optimum 2 was established in the preceding section. For expected-cost optimality, the finite recursion (3) is an exact Bellman equation: conditional on a residue y, each $h \in S_{a,y}$ is equiprobable, since z_i is uniform on E and $y = ah + z_i$ has one admissible noise value per candidate. Thus $\Pr(y \mid S, a) = |S_{a,y}|/(5|S|)$. Exhausting the 16 nonzero slopes at each of the three slots gives the following eight pairs of symmetric first slopes a and $17 - a$, with the optimum over all remaining decisions:

a (and $17 - a$)	1	2	3	4	5	6	7	8
expected pulses	199/125	9/5	44/25	219/125	209/125	237/125	44/25	206/125

The option of skipping the first slot has exact value 44/25. Consequently the global three-slot minimum is 199/125.

Here is an explicit optimal decision tree. Use normalized slope $a = 1$ (rank-four pulse amplitude 7) at time 14. If the first observation y leaves a singleton, stop. The ambiguous posterior sets and a valid rank-five second slope at time 15 are

y	$S_{1,y}$	second slope a_2
3	$\{3, 16\}$	2
5	$\{3, 5\}$	3
6, 7	$\{3, 5, 6\}$	4
8, 9	$\{5, 6\}$	5

All pairs are resolved by the second pulse. For a triple $S = \{3, 5, 6\}$, slope 4 resolves every second observation except $y_2 = 7$, which leaves $\{5, 6\}$; there a final rank-six slope 5 resolves the pair. Actual pulse amplitudes are $7a_i \pmod{17}$. The first observation determines a singleton in 11 of the 25 equally likely (h, z_{14}) combinations. Of the 14 remaining combinations, exactly 4/125 of all source/noise trajectories require a third pulse. Therefore the expected number is

$$1 + \frac{14}{25} + \frac{4}{125} = \frac{199}{125}.$$

For the stricter two-pulse budget, the same exhaustive finite Bellman calculation with the pulse budget as an additional state finds a minimum 44/25 (for example, with first slope 7 or 10). The result is a finite exact certificate, not an asymptotic theorem. The finite calculation uses exact rational probabilities and exhaustively enumerates the actions, residues and posterior candidate sets. The resulting policies were also checked directly against all $5 \cdot 5^3 = 625$ possible CCR–10 source/noise trajectories. These finite checks follow from the stated Bellman recursion and source alphabets. □

Remark 5.2 (Interpretation)**.** *The exact cost gap is a property of the* specified *finite source alphabet, independent uniform diagnostic nuisance, available rank–time slots and pulse-count cost. It is not a universal probability law for CCR–k, nor does it show that adaptive decision trees are novel. The new check distinguishes a worst-case-optimal policy from an expected-cost-optimal policy, even within this one CCR–10 realization.*

6 A finite-field criterion and a cross-prime family

The previous results used $q = 17$. We now isolate the exact finite-field condition behind the adaptive separation and verify its CCR realization at several different prime moduli. Let $q > 16$ be prime and let $H, E \subset \mathbb{F}_q$ be nonempty finite sets. At a nonzero normalized slope a, a controlled measurement returns

$$y = ah + z, \qquad h \in H, \quad z \in E.$$

The same unknown h persists, whereas z may be chosen independently and adversarially on each used opportunity. Write

$$D = E - E, \quad S_{a,y} = \{h \in H : y - ah \in E\}, \quad \mathcal{B}(S) = \{b \in \mathbb{F}_q^\times : b((S - S) \setminus \{0\}) \cap D = \varnothing\}.$$

Theorem 6.1 (Exact two-stage criteria)**.** *A predetermined two-measurement design exists if and only if there are $a, b \in \mathbb{F}_q^\times$ such that*

$$((a^{-1}D) \cap (b^{-1}D) \cap ((H - H) \setminus \{0\})) = \varnothing.$$

An adaptive controller using at most two measurements identifies h for all allowed nuisance values if and only if there is $a \in \mathbb{F}_q^\times$ such that

$$\mathcal{B}(S_{a,y}) \neq \varnothing \quad \text{for every } y \text{ with } |S_{a,y}| \geq 2.$$

Thus a strict adaptive/predetermined worst-case gap at two measurements occurs exactly when the second criterion holds and the first fails, provided no one-measurement design succeeds. If a predetermined three-measurement design exists, its optimum is then exactly three.

Proof. For a pair $h \neq h'$, the observation sets $ah + E$ and $ah' + E$ intersect if and only if $a(h - h') \in E - E = D$. Two fixed observations cannot distinguish that pair precisely when both scaled differences belong to D, proving the first equivalence. Conditional on an observed first residue y, the exact remaining candidate set is $S_{a,y}$. One more measurement can separate all candidates precisely when a nonzero slope b avoids D on all of its nonzero pairwise differences, which means $b \in \mathcal{B}(S_{a,y})$. Choosing such a b after seeing y proves both necessity and sufficiency. The last statement follows by comparison of the two criteria and the existence of a three-test witness. □

Proposition 6.2 (Exact nonresetting CCR–10 embedding)**.** *Fix $p = 5$, $k = 10$, an old source $h = a_{4,4} \in H$, and prime $q > 16$. The high sources at times $11, 12, 13$ can be chosen, depending on h, to make all three full coefficient rows independent of h. At rank–time opportunities $(14, 4), (15, 5), (16, 6)$, fix all fresh high sources to zero except a controlled nonzero source of rank r, a diagnostic nuisance source of rank $r + 4$ in E, and optionally unrestricted sources above rank $r + 4$. Then the normalized diagnostic is exactly*

$$w_r = c_q \alpha_r h + z_r, \qquad c_q = (3)_3[(10)_4]^{-1} \in \mathbb{F}_q^\times.$$

Every slope in $\mathbb{F}_q^\times$ is attainable. Thus the criteria in Theorem 6.1 apply to a single CCR–10 system without resetting between observations. If two hypotheses have identical normalized diagnostics, unrestricted higher sources can match the remaining entries of the entire observed row by chronological triangular elimination. The claim concerns this specified source and observation contract, not arbitrary unknown CCR sources.

Proof. The diagonal coefficient of fresh source $a_{j,n}$ in coefficient $C_{n,j}$ equals $(n-1)_{j-1}$. The needed factors at times 11–16 and degrees at most ten are units modulo any prime $q > 16$. Consequently fresh ranks $4, \ldots, 10$ at each masking time can be solved successively to reproduce any given reference row. At the opportunity $s = 10 + r$, the previous row indexed $s - r$ is row ten, and its degree-four coefficient is $B + (3)_3 h$, with B independent of h. Subtracting the known three-lag baseline and the controlled contribution $(s-1)_{r-1}\alpha_r B$, then dividing by $(s-1)_{r+3}$, gives $c_q \alpha_r h + z_r$. The free sources of greater rank can then match subsequent coefficients without altering the diagnostic, by the same triangular argument. □

Proposition 6.3 (Affine invariance)**.** *For nonzero $u, w \in \mathbb{F}_q$ and any $v, d \in \mathbb{F}_q$, replacing H by $uH + v$ and E by $wE + d$ preserves the optimal adaptive and predetermined intervention counts. Under corresponding uniform priors and nuisance laws, it preserves the optimal expected intervention counts as well.*

Proof. Under the substitution $h' = uh + v$, $z' = wz + d$, choose $a' = wa/u$. Then $a'h' + z' = w(ah + z) + a'v + d$. The two sides have mutually invertible, known output changes and the choice $a \leftrightarrow a'$ is a bijection on nonzero slopes. Thus every fixed or adaptive decision tree transports bijectively, including its probability law and cost. □

Proposition 6.4 (Four certified prime instances)**.** *For each row of the following table, with $E = \{0, \ldots, b - 1\}$, there is a two-pulse adaptive strategy; every predetermined two-pulse*

schedule fails; and a predetermined three-pulse strategy succeeds. All instances are realized by Proposition 6.2.

q	H	b	*first adaptive slope*	*fixed triple*
17	$\{3, 5, 6, 11, 16\}$	5	2	$(1, 2, 5)$
19	$\{2, 5, 7, 17, 18\}$	5	1	$(1, 2, 5)$
23	$\{0, 10, 13, 16, 18, 22\}$	5	1	$(1, 2, 5)$
29	$\{8, 10, 18, 19, 21, 25\}$	7	1	$(1, 2, 7)$

The exact optimal expected pulse counts with three opportunities under uniform h and independent uniform nuisances are respectively $199/125, 199/125, 22/15, 11/7$. *If worst-case pulse count is bounded by two, the expected optima are* $44/25, 42/25, 22/15, 11/7$.

Proof. The first slope in each row satisfies the second condition of Theorem 6.1, as verified by its finite table of nonempty posterior sets and their separating slopes. For every ordered nonzero slope pair, the first condition fails: exhaustive finite enumeration of the $(q-1)^2$ slope pairs produces a colliding source pair for each. The displayed fixed triple passes the same exact pair-difference test. Since $|H||E| > q$ in each case, one test cannot separate all h by the pigeonhole principle. Exact rational Bellman recursions over all 3 available opportunities, including skipped slots, give the expected costs; complete source/diagnostic trajectories are independently checked in the CCR–10 recurrence. This is a finite reproducible certificate for four concrete instances, not an assertion for all primes. □

Proposition 6.5 (Large-prime obstruction for fixed alphabets)**.** *Write* $\Delta = ((H - H) \setminus \{0\})$ *and* $D = E - E$. *If*

$$q - 1 > |\Delta|\,(|D| - 1),$$

then some nonzero normalized slope identifies h with a single measurement despite arbitrary $z \in E$. In particular, for any fixed finite sets of distinct integers H, E reduced modulo primes q sufficiently large, the adaptive-versus-predetermined pulse-count gap must disappear. More explicitly it suffices (when residues remain distinct) that

$$q - 1 > |H|(|H| - 1)|E|(|E| - 1).$$

Proof. For each nonzero old-source difference $d \in \Delta$, a slope a causes a collision only if $ad \in D \setminus \{0\}$. Since d is invertible over $\mathbb{F}_q$, there are exactly $|D| - 1$ bad slopes for this d. The union over d contains at most $|\Delta|(|D| - 1)$ slopes. If this is smaller than $q - 1$, at least one nonzero slope avoids every collision. Finally $|\Delta| \le |H|(|H| - 1)$ and $|D| - 1 \le |E|(|E| - 1)$, yielding the stated uniform bound. □

Remark 6.6 (Interpretation)**.** *The difference-set and decision-tree equivalences are standard finite identification tools. The CCR-specific construction consists of their exact nonresetting embedding in the CCR chronology together with multiple prime-field instances and explicit costs. Affine changes create isomorphic instances, not fundamentally different arithmetic theories. The table does not establish an asymptotic density of suitable primes.*

7 An explicit infinite-prime adaptive gap

The preceding finite examples may be replaced by a genuine arithmetic family. The difference-set argument below is elementary; its significance here is the realization of a fixed-versus-adaptive gap at *every sufficiently large prime characteristic* inside a nonresetting CCR–10 chronology with the specified restricted-input observation contract.

For each prime $q \geq 41$, let

$$H = \{0, 2, 7, 8\} \subset \mathbb{F}_q, \qquad E_q = \{0, 1, \ldots, \lfloor q/3 \rfloor\} \subset \mathbb{F}_q.$$

A pulse of nonzero normalized slope a produces $y = ah+z \pmod q$ with $h \in H$ fixed throughout the experiment and a new independent adversarial $z \in E_q$ at each pulse. We analyze these normalized diagnostic residues first. Under the permitted unrestricted higher-rank nuisance inputs, the subsequent full-row lifting lemma proves that additional coefficient entries do not weaken the robust lower bound. For $h \neq h'$, the corresponding possible diagnostic outputs overlap exactly when

$$a(h - h') \in E_q - E_q = \{t \in \mathbb{F}_q : \|t/q\| \leq \lfloor q/3 \rfloor / q\}.$$

Here $\|x\|$ denotes distance to the nearest integer. Because $q > 3$ is prime, none of the nonzero residues ad with $d \in \{1, 2, 5, 6, 7, 8\}$ lies on a boundary $\|ad/q\| = 1/3$.

Lemma 7.1 (Circle difference certificate)**.** *Let $\mathcal{D} = \{1, 2, 5, 6, 7, 8\}$, and for $x \in \mathbb{R}/\mathbb{Z}$ away from the finitely many threshold points define*

$$B(x) = \{d \in \mathcal{D} : \|dx\| < 1/3\}.$$

Then $B(x) \cap B(y) \neq \varnothing$ for all such x, y.

Proof. By the symmetry $B(x) = B(1 - x)$, it suffices to inspect $0 < x < 1/2$. The indicator of each d changes only at $x = (j \pm 1/3)/d$. Sorting all these rational breakpoints for $d \in \mathcal{D}$ produces 25 open subintervals of $(0, 1/2)$. Direct evaluation at one rational midpoint of each subinterval shows that every possible $B(x)$ contains at least one of the following seven sets:

$$\{1, 2\}, \quad \{1, 5, 6\}, \quad \{1, 5, 8\}, \quad \{1, 6, 7\},$$
$$\{1, 7, 8\}, \quad \{2, 5, 7\}, \quad \{2, 6, 8\}.$$

These seven sets are pairwise intersecting (which is checked immediately from the displayed list). Hence every two possible collision sets intersect. The precise rational breakpoint partition and an exact-integer checker are included with the accompanying reproducibility archive; the statement for prime residues does not require evaluating threshold points. □

Theorem 7.2 (Infinite-prime sharp adaptive separation)**.** *For every prime $q \geq 41$ in the above diagnostic experiment:*

1. *One pulse cannot identify h for every nuisance value.*
2. *No two predetermined nonzero slopes identify h for every pair of nuisance values.*
3. *There exists a predetermined three-pulse schedule that identifies h.*
4. *An adaptive strategy identifies h using at most two pulses.*

Therefore the optimal worst-case pulse counts are exactly three for predetermined schedules and exactly two for adaptive schedules. The same experiment is realizable without resetting between pulses in the CCR–10 source chronology at times $14, 15, 16$, under the explicit source and readout restrictions of the previous chronological-embedding theorem.

Proof. (1) If one pulse separated every source, the four sets $ah + E_q$ would be pairwise disjoint in $\mathbb{F}_q$. But $4|E_q| = 4(\lfloor q/3 \rfloor + 1) > q$.

(2) The positive source differences are precisely $\mathcal{D}$. For any two fixed slopes a, b, the collision sets $B(a/q)$ and $B(b/q)$ intersect by Lemma 7.1. Thus there is a nonzero source difference

$d \in \mathcal{D}$ that can be masked by nuisance differences at both interventions. Independent nuisances produce a common diagnostic pair, so no fixed pair succeeds.

(3) Set $a_1 = \mathrm{nint}(3q/40)$, $a_2 = \mathrm{nint}(q/5)$ and $a_3 = \mathrm{nint}(3q/7)$, where nint denotes nearest-integer rounding. For every prime $q \geq 61$ the fractions a_i/q lie respectively in the open intervals

$$(1/15, 1/12), \qquad (4/21, 5/24), \qquad (5/12, 4/9),$$

because their target fractions lie in these intervals with margin at least $1/120$, and nearest rounding has error at most $1/(2q)$. The three corresponding collision patterns are $\{1, 2\}$, $\{1, 5, 6\}$ and $\{2, 5, 7\}$, whose intersection is empty. For the five smaller eligible primes $41, 43, 47, 53, 59$, direct residue checks with the same formulas likewise give an empty triple intersection. Consequently the three measurements distinguish every pair of hidden sources.

(4) Choose the first slope $a = \mathrm{nint}(5q/16)$. The four circular interval start points $ah/q \pmod 1$ for $h \in H$ approximate $0, 3/16, 1/2, 5/8$. Every three of the four ideal start points require a circular arc of length at least $7/16$ to contain them. Nearest rounding changes the circular separation of any pair by at most $4/q$ (the diameter of H is eight). Thus for $q \geq 41$ no three actual start points fit inside an arc of length $1/3$, since $7/16 - 4/q > 1/3$. As every translate $ah + E_q$ is a circular interval of length at most $1/3$, each first residue y leaves at most two candidates. If only one remains, stop. Otherwise let the two candidates have nonzero difference $d = h - h'$. Choose the second slope

$$b = \frac{q-1}{2}\, d^{-1} \pmod q.$$

Then $bd \equiv (q-1)/2 \pmod q$, which does not belong to $E_q - E_q$ since $\lfloor q/3 \rfloor < (q-1)/2$ for $q \geq 41$. The second observation consequently separates the pair even under an adversarial fresh nuisance.

Finally, for $p = 5, k = 10$ the source coefficients at times $11, 12, 13$ can be chosen recursively by the invertible triangular pivots $(n-1)_{r-1}$ to prepare identical full output rows for every $h \in H$. At times $14, 15, 16$ use rank–$4, 5, 6$ pulses with the predetermined or adaptively chosen normalized slopes, and independent nuisance inputs at ranks $8, 9, 10$, respectively. The normalization of the selected rank–$(r+4)$ diagnostic is $y = ah + z$, with all required denominators invertible because $q \geq 41 > 16$. This is the chronological embedding established before this section, and requires no system reset. □

Remark 7.3 (Interpretation)**.** *This theorem is an infinite family, unlike an affine relabeling of one finite example. The source alphabet H remains fixed, but the nuisance alphabet E_q grows linearly with q, so it does not contradict the previous eventual-one-pulse bound for* both *alphabets fixed. The circle difference-set calculation and adaptive elimination are classical-style combinatorial arguments, not standalone priority claims. The CCR-specific result is the exact nonresetting chronological realization under a transparent restricted diagnostic observation contract. Under the robust source contract of Section 8, additional complete coefficient rows do not remove the lower bound; this conclusion is not asserted for other source contracts or for disclosure of the unmasked pre-preparation history.*

8 Full-row robust observability under restricted fresh-source freedom

The distinction between a scalar diagnostic and disclosure of the full CCR coefficient row requires care. We record an explicit lifting statement. The observer is shown the *prepared rows from time 11 onward*, not the unmasked rows at times $0, \ldots, 10$; the value of the hidden old

source is not disclosed during preparation. These observation-window restrictions are part of the experiment, not consequences of the CCR recurrence.

Fix $p = 5$, $k = 10$ and a prime $q \geq 41$, and let $h = a_{4,4} \in H = \{0, 2, 7, 8\}$. All high sources prior to time 11 apart from h vanish. The fresh ranks $4, \ldots, 10$ at times $11, 12, 13$ may be chosen depending on h so that the three entire coefficient rows are identical for all $h \in H$. The observer knows the rule defining the admissible family, but observes the resulting rows rather than the source choices.

There are three designated pulse opportunities $(s, r) = (14, 4), (15, 5), (16, 6)$. At a selected opportunity the controlled rank-r source has a known nonzero amplitude α, the unknown rank-$(r + 4)$ source lies in $E_q = \{0, \ldots, \lfloor q/3 \rfloor\}$, all other new source ranks below $r + 4$ vanish, and ranks strictly above $r + 4$ may take unrestricted values in $\mathbb{F}_q$. If an opportunity is skipped, every fresh high source at that time is zero. This is the *robust full-row source contract*.

Lemma 8.1 (Lifting scalar collisions to entire CCR rows)**.** *Suppose two distinct old sources $h, h' \in H$ have produced the same three prepared rows. At one designated pulse opportunity with controlled normalized slope*

$$a = c\alpha, \qquad c = (3)_3 \left[(10)_4\right]^{-1} \in \mathbb{F}_q^*,$$

the two histories can produce the same entire *current coefficient row with permitted fresh nuisance inputs if and only if*

$$a(h - h') \in E_q - E_q.$$

If they produce a common row, the same assertion may be applied at every subsequent opportunity without restarting either history. Skipped opportunities preserve equality of rows.

Proof. Write $s = 10 + r$ with $r \in \{4, 5, 6\}$. The contribution to the current row not involving fresh high sources is identical because the preceding three full rows agree. The chosen rank-r input contributes $(s - 1)_{r-1} \alpha u^r P_{10}$; the old histories have identical coefficients of P_{10} below degree four and

$$C_{10,4}(h) - C_{10,4}(h') = (3)_3 (h - h').$$

Consequently, all current coefficients below degree $r + 4$ agree before choosing a rank-$(r + 4)$ input. In degree $r + 4$, equality holds exactly if

$$(s - 1)_{r-1} \alpha (3)_3 (h - h') + (s - 1)_{r+3} (z - z') = 0 \pmod{q}$$

for some $z, z' \in E_q$. Since $(s - 1)_{r+3} = (s - 1)_{r-1} (s - r)_4$ and $s - r = 10$, this condition is exactly $c\alpha(h - h') \in E_q - E_q$ (the sign is immaterial because $E_q - E_q$ is symmetric).

At each higher degree $j > r + 4$, the fresh rank-j source has coefficient $(s - 1)_{j-1}$ in the j-th observed coefficient. This is a unit in $\mathbb{F}_q$, because $s \leq 16 < q$. Thus, proceeding from degree $r + 5$ to 10, choose the unrestricted source in the second history to match the first history's current coefficient, without disturbing lower degrees. Both entire rows now agree. The argument also proves necessity, since no source of rank higher than $r + 4$ can affect degree $r + 4$.

At a skipped time, all fresh high sources are zero and the CCR recurrence only uses the preceding three full rows, which are identical; so equality persists. This supplies induction through all three opportunities. □

Corollary 8.2 (Sharp full-row gap for the infinite prime family)**.** *For every prime $q \geq 41$, in this source contract, observing* all *coefficient rows at times $11, 12, \ldots, 16$ does not reduce the worst-case number of predetermined pulse opportunities below three. An adaptive strategy needs at most two pulses even with the full rows disclosed, and a single pulse cannot succeed. Thus the worst-case optima remain three for predetermined schedules and two for adaptive schedules.*

Proof. The scalar circle-difference theorem in Section 7 gives a common source difference for any two predetermined normalized pulse slopes. The lemma lifts the two scalar collisions to identical complete coefficient rows and propagates equality through skipped time slots. It also lifts the one-pulse pigeonhole collision. Conversely, the scalar projection of a complete row is available to the observer, so the existing adaptive two-pulse construction and predetermined three-pulse construction retain their upper bounds. All preparation rows were equalized before the first pulse. □

Remark 8.3 (Source-contract dependence)**.** *No lower bound is claimed if the unmasked rows before time 11, the hidden source inputs themselves, or the unrestricted high-rank nuisance values are revealed, or if the source contract forbids choosing higher-degree nuisances to equalize the rows. The controller may choose its next pulse using the previously observed full coefficient rows, but all comparisons are between histories with the same observable prefix. The theorem is therefore a robust identifiability result for the explicitly allowed input family, not an unconditional claim about every CCR system.*

8.1 Exact rational certificate for the circle lemma

For completeness, the 25 open cells on $0 < x < 1/2$ and their full collision sets $B(x)$ are listed below. This table is obtained by sorting all rational boundaries $(j \pm 1/3)/d$ for $d \in \{1, 2, 5, 6, 7, 8\}$. Each row can be verified using the midpoint of the indicated interval. No residue a/q for primes $q \geq 41$ can be one of these boundaries, since $3ad = 3jq \pm q$ would force $q \mid d$.

No.	Interval	Collision set
1	$(0, 1/24)$	$\{1, 2, 5, 6, 7, 8\}$
2	$(1/24, 1/21)$	$\{1, 2, 5, 6, 7\}$
3	$(1/21, 1/18)$	$\{1, 2, 5, 6\}$
4	$(1/18, 1/15)$	$\{1, 2, 5\}$
5	$(1/15, 1/12)$	$\{1, 2\}$
6	$(1/12, 2/21)$	$\{1, 2, 8\}$
7	$(2/21, 1/9)$	$\{1, 2, 7, 8\}$
8	$(1/9, 2/15)$	$\{1, 2, 6, 7, 8\}$
9	$(2/15, 1/6)$	$\{1, 2, 5, 6, 7, 8\}$
10	$(1/6, 4/21)$	$\{1, 5, 6, 7\}$
11	$(4/21, 5/24)$	$\{1, 5, 6\}$
12	$(5/24, 2/9)$	$\{1, 5, 6, 8\}$
13	$(2/9, 5/21)$	$\{1, 5, 8\}$
14	$(5/21, 4/15)$	$\{1, 5, 7, 8\}$
15	$(4/15, 5/18)$	$\{1, 7, 8\}$
16	$(5/18, 7/24)$	$\{1, 6, 7, 8\}$
17	$(7/24, 1/3)$	$\{1, 6, 7\}$
18	$(1/3, 8/21)$	$\{2, 5, 6, 8\}$
19	$(8/21, 7/18)$	$\{2, 5, 6, 7, 8\}$
20	$(7/18, 5/12)$	$\{2, 5, 7, 8\}$
21	$(5/12, 4/9)$	$\{2, 5, 7\}$
22	$(4/9, 11/24)$	$\{2, 5, 6, 7\}$
23	$(11/24, 7/15)$	$\{2, 5, 6, 7, 8\}$
24	$(7/15, 10/21)$	$\{2, 6, 7, 8\}$
25	$(10/21, 1/2)$	$\{2, 6, 8\}$

Finite verification. Exact arithmetic checks cover all 625 source–nuisance trajectories and 256 nonzero slope pairs for $q = 17$, and all 16,440 two-slot schedules over $q = 41, 43, 47$. These finite calculations corroborate the constructions; the infinite-prime conclusion follows from the proofs above.

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